



Ethical Considerations in Deep Learning

Project 4- Report

Course: Deep Learning

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Date: 04/09/2026

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Introduction

Deep Learning systems now shape decisions in hiring, lending, healthcare, criminal justice, and everyday digital services. Their predictive power comes from learning patterns in large datasets, but this same process can introduce serious ethical risks. This report identifies three key ethical issues in Deep Learning, examines a real-world case where such a system caused concern, and proposes practical measures to reduce these risks.

Task 1: Three Ethical Issues in Deep Learning

1. Bias and Discrimination

Deep Learning models learn from historical training data. If that data reflects existing social biases or under-represents certain groups, the model reproduces and can even amplify those biases in its predictions. This has been observed in hiring tools, facial recognition systems, and credit-scoring algorithms, where outcomes differ unfairly across gender, race, or age groups.

2. Privacy Violation

Training effective Deep Learning models typically requires massive volumes of data, which often includes personal or sensitive information such as facial images, medical records, browsing behavior, or financial data. This data is frequently collected without explicit, informed consent, and models themselves can sometimes be probed to reveal information about individuals in the training set, raising serious privacy concerns.

3. Lack of Transparency (the “Black Box” Problem)

Many Deep Learning architectures, particularly deep neural networks, are so complex that even their developers cannot fully explain why a specific decision was made. This opacity undermines accountability, makes errors difficult to detect and correct, and is especially problematic when such systems are used in high-stakes domains like healthcare, criminal justice, or financial approval.

Task 2: Real-World Example

Case Study: The COMPAS Recidivism Risk Algorithm

COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is an algorithmic tool used by courts in parts of the United States to predict the likelihood that a defendant will reoffend. Judges have used its risk scores to help inform decisions on bail, sentencing, and parole.

In 2016, an investigation by the news organization ProPublica analyzed COMPAS risk scores against actual reoffending outcomes. The analysis found that Black defendants were disproportionately labeled as “higher risk” compared with white defendants, even when their actual likelihood of reoffending was similar. White defendants, by contrast, were more often incorrectly labeled as lower risk.

This case illustrates two of the issues identified above at once: algorithmic bias, because the model's predictions varied unfairly by race, and lack of transparency, because the company that developed COMPAS treated the algorithm as proprietary and refused to disclose how it calculated its risk scores — making independent verification and correction difficult.

Task 3: Suggested Measures

Addressing these ethical issues requires action at the level of data, algorithms, oversight, and policy.

Reducing Bias

- Use diverse, representative training datasets that include balanced coverage of gender, race, age, and other relevant groups.
- Regularly audit models for disparate impact across demographic groups before and after deployment.

Protecting Privacy

- Apply data minimization and anonymization techniques, and obtain informed consent before collecting personal data.
- Use privacy-preserving methods such as federated learning or differential privacy when training on sensitive data.

Improving Transparency and Accountability

- Adopt Explainable AI (XAI) techniques so model decisions can be interpreted rather than treated as a black box.
- Keep a human decision-maker in the loop for high-stakes outcomes rather than relying solely on automated scores.
- Support regulation and independent auditing, such as the EU AI Act or similar frameworks, requiring disclosure and testing of high-risk AI systems.

Conclusion

Deep Learning offers significant benefits, but its risks of bias, privacy violation, and opacity are not hypothetical — they have already caused real harm, as the COMPAS case demonstrates. Building more diverse datasets, applying explainability techniques, maintaining human oversight, and supporting sound regulation are practical steps that can help ensure Deep Learning systems are deployed responsibly and fairly.